

Variance Component Estimation from Nuclear Family Data: Theory, Mixed Model Implementation, and Power Analysis

Abstract

Quantitative genetic research traditionally relies on twin studies to partition phenotypic variation into additive genetic (A), dominance genetic (D), shared environmental (C), and unique environmental (E) components. However, many contemporary cohort and biobank datasets consist primarily of nuclear families rather than twins.

This paper presents a unified likelihood-based framework for variance component estimation using nuclear family data. We show that nuclear family trios identify additive genetic effects and a single familial resemblance component but cannot separate shared environmental and dominance genetic effects. We then demonstrate that inclusion of siblings provides sufficient covariance information to identify the full ACDE model.

A practical implementation in **R** using linear mixed models is provided together with simulation studies verifying unbiased estimation. A Monte Carlo power analysis quantifies the sample sizes required to detect shared environmental and dominance effects.

This framework enables rigorous quantitative genetic analysis in modern family-based cohorts and provides guidance for study design.

1 Introduction

Biometrical genetics partitions phenotypic variance into

$$V_P = V_A + V_C + V_D + V_E$$

where A denotes additive genetic effects, D dominance effects, C shared environment, and E unique environment.

Twin designs have historically dominated this field, but large population cohorts increasingly contain nuclear family data. This paper develops a unified framework for variance decomposition using nuclear families.

2 General Pedigree Covariance

For individuals i and k ,

$$\text{Cov}(y_i, y_k) = 2\phi_{ik}V_A + \Delta_{ik}V_D + \gamma_{ik}V_C$$

- ϕ_{ik} kinship coefficient
- Δ_{ik} probability of sharing two alleles IBD
- γ_{ik} shared environment indicator

3 Identifiability in Nuclear Family Trios

Expected covariances:

Pair	Expected covariance
Parent–parent	V_C
Parent–child	$\frac{1}{2}V_A + V_C$

Dominance effects are never shared between parents and offspring:

$$\text{Cov}_D(\text{parent}, \text{child}) = 0$$

Hence only

$$\text{Cov}_{family} = \frac{1}{2}V_A + V_F$$

is identifiable, where V_F is a generic familial resemblance component. Therefore trios allow AE, ACE or ADE models but not ACDE.

4 Why Siblings Enable ACDE

Sibling covariance:

$$\text{Cov}_{sib} = \frac{1}{2}V_A + V_C + \frac{1}{4}V_D$$

This additional $\frac{1}{4}V_D$ term separates C and D .

Design	Identifiable components
Trio	AE, ACE or ADE
Families with siblings	ACDE

5 Likelihood Formulation

For family phenotype vector $\mathbf{y}_i$:

$$\mathbf{y}_i \sim \mathcal{N}(0, \Sigma)$$

Log-likelihood:

$$\ell = -\frac{1}{2} \sum_i [\log |\Sigma| + \mathbf{y}_i^T \Sigma^{-1} \mathbf{y}_i + k \log(2\pi)]$$

Variance parameterisation:

$$V_A = e^{\theta_A}, V_C = e^{\theta_C}, V_D = e^{\theta_D}, V_E = e^{\theta_E}$$

6 Mixed Model Representation

$$\mathbf{y} = \mathbf{X}\beta + Z_A a + Z_C c + Z_D d + \varepsilon$$

$$a \sim N(0, V_A), c \sim N(0, V_C), d \sim N(0, V_D), \varepsilon \sim N(0, V_E)$$

7 R Implementation (nlme)

```
library(nlme)
```

```
fit_ACDE <- lme(
  y ~ 1,
  random = list(
    famid = pdBlocked(list(
      pdDiag(~ Acoef - 1),
      pdDiag(~ Ccoef - 1),
      pdDiag(~ Dcoef - 1)
```

```

    )),
    indid = pdDiag(~ 1)
  ),
  data = longdata,
  method = "REML"
)

```

8 Simulation Study

Data were simulated under known parameters to verify unbiased recovery. Across repeated simulations the model recovered V_A, V_C, V_D, V_E without systematic bias.

9 From Simulation to Power Analysis

Although unbiased, detectability depends strongly on sample size. We therefore performed Monte Carlo power analyses using likelihood ratio tests for V_C and V_D .

10 Power Analysis Results

Key findings:

- V_A detectable with moderate samples.
- Detecting V_C requires large cohorts.
- Detecting V_D requires very large cohorts.

11 Discussion

Nuclear family cohorts provide a practical alternative to twin designs. However, the full ACDE model requires siblings and large sample sizes. Dominance and shared environment are statistically demanding to detect.

12 Conclusion

Nuclear family data enable rigorous variance decomposition without twins. Trio data estimate additive genetics and familial resemblance; families with siblings enable full ACDE estimation.

Power analyses show that large cohorts are essential for detecting shared environmental and dominance effects, providing critical guidance for study design in modern population genetics.